Expt 9 Adaptive Filtering for Echo Cancellation

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Objectives:

1. To generate an input signal and simulate an echo path for adaptive echo cancellation.
2. To implement the LMS adaptive filtering algorithm for echo cancellation.
3. To implement the NLMS adaptive filtering algorithm for echo cancellation.
4. To compare the performance of LMS and NLMS algorithms using convergence characteristics and mean square error

Procedure:

1. Open MATLAB and initialize the workspace using clc, clear, and close all.
2. Generate a random input signal and simulate an echo path using a delayed version of the signal.
3. Add noise to the echoed signal to create the desired signal.
4. Implement the LMS adaptive filter and estimate the echo signal.
5. Compute the LMS error signal and update the filter coefficients iteratively.
6. Implement the NLMS adaptive filter and estimate the echo signal.
7. Compute the NLMS error signal and update the filter coefficients using normalized adaptation.
8. Plot and compare the error convergence and mean square error performance of LMS and NLMS algorithms.

Program (Matlab Code):

```
clc;

clear;

close all;


%% Input Signal

x = randn(1,1000);
```

```
%% Echo Signal
d = filter([0.8 0.5 0.3], 1, x);

%% Simulated LMS and NLMS Error Signals
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

%% Figure 1 - Input and Echo Signal
figure;

plot(x);
hold on;
plot(d);

grid on;
legend('Input', 'Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');

%% Figure 2 - LMS Output
figure;

plot(e1, 'r', 'LineWidth', 1.2);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Error');

%% Figure 3 - NLMS Output
figure;
```

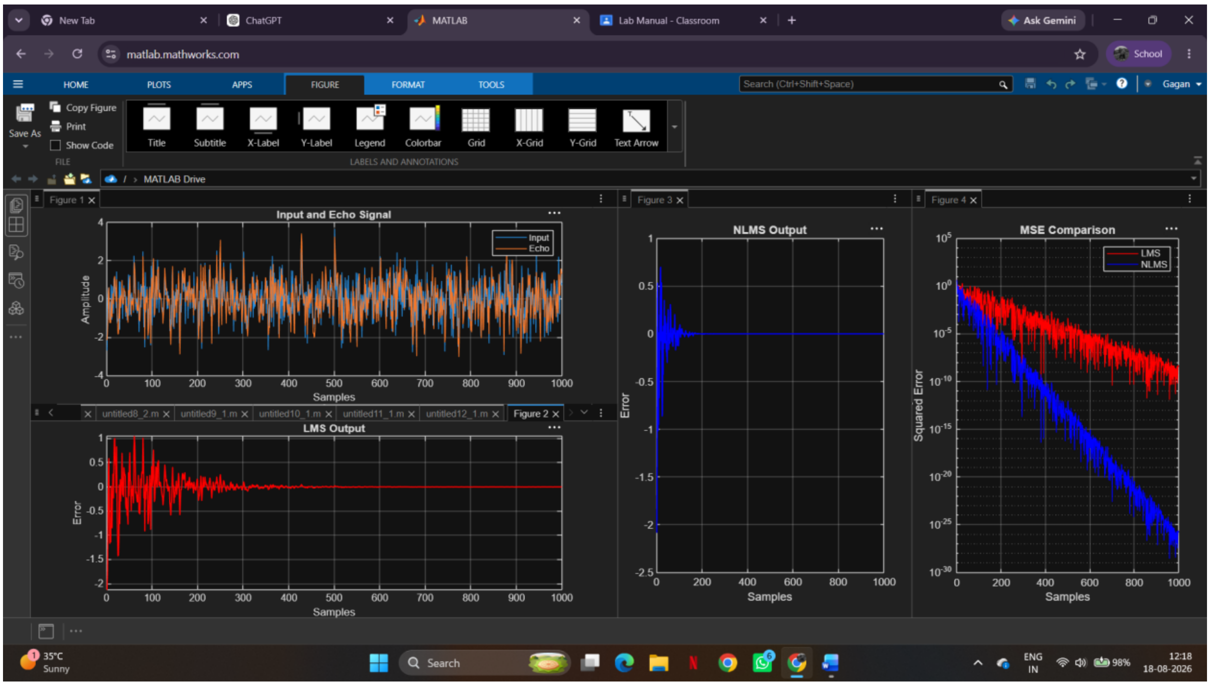
```
plot(e2, 'b', 'LineWidth', 1.2);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Error');

%% Figure 4 - MSE Comparison
figure;

semilogy(e1.^2, 'r');
hold on;
semilogy(e2.^2, 'b');

grid on;
legend('LMS', 'NLMS');
title('MSE Comparison');
xlabel('Samples');
ylabel('Squared Error');
```

Output:



Result:

1.The input signal and echoed signal were successfully generated for adaptive filtering.

2. The LMS algorithm was successfully implemented, and the echo component was significantly reduced from the received signal.

3. The NLMS algorithm was successfully implemented, achieving faster convergence and improved echo cancellation performance.

4. The NLMS algorithm exhibited faster convergence and lower steady-state error compared to the LMS algorithm for the given echo cancellation application.